

# Project report

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*Data Preparation, Backtesting, and Performance Evaluation*

**Term Project — Data Science in Finance (MSDS 451)**

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Repository: [github.com/nadishiro17/nadia-etf](https://github.com/nadishiro17/nadia-etf)

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## Executive Summary

This report covers the empirical phase of the term project: data acquisition (Checkpoint B) and backtesting / performance evaluation (Checkpoint C). The proposed fund is a three-book barbell ETF combining a defensive mean-reversion book, a growth momentum book, and a low-volatility defensive book, allocated by risk parity with a rule-based regime tilt.

Over the 26-year historical period (1999–2024) and across 500 Monte Carlo synthetic histories, the strategy produces:

| Metric            | Historical | Walk-forward OOS | MC median | SPY actual |
|-------------------|------------|------------------|-----------|------------|
| CAGR              | 12.02%     | 11.03%           | 13.4%     | 8.11%      |
| Annual volatility | 15.51%     | 14.91%           | —         | 19.34%     |
| Sharpe ratio      | 0.62       | 0.58             | 0.65      | 0.35       |
| Max drawdown      | −36%       | −36%             | −30%      | −55%       |
| Annualized alpha  | +5.25%     | +3.38%           | —         | 0%         |
| Beta              | 0.65       | 0.66             | —         | 1.00       |

The strategy beats SPY in 97.4% of Monte Carlo paths and maintains a max drawdown better than −50% in 98.6% of paths. At a realistic fee structure of 1% management + 15% performance fee on excess over SPY, the net CAGR remains 9.93% — still ~180 bps above passive SPY ownership. This is a viable business proposition for an actively managed ETF.

## 1. Data (Checkpoint B)

### 1.1 Sources

| Series                | Source                               | Frequency               | Coverage                 |
|-----------------------|--------------------------------------|-------------------------|--------------------------|
| Stock prices          | Yahoo Finance via yfinance ≥0.2.40   | Daily OHLCV + Adj Close | 1999-01-04 to 2024-12-30 |
| Benchmark (primary)   | SPY ETF — total return via Adj Close | Daily                   | Same                     |
| Benchmark (reference) | ^GSPC — S&P 500 price index          | Daily                   | Same                     |
| Risk-free rate        | ^IRX — 13-week T-bill yield          | Daily                   | Same                     |

Why SPY rather than ^GSPC as primary benchmark: SPY's Adjusted Close reinvests dividends, making it a true total return index comparable to our 50 stocks' adjusted closes. The raw S&P 500 price index (^GSPC) understates returns by ~2 pp/year because it excludes dividends — using it would artificially inflate apparent alpha.

## 1.2 Securities universe

50 large-cap US equities with continuous price history since January 4, 1999, diversified across 10 GICS sectors:

| Sector                 | n | Tickers                                      |
|------------------------|---|----------------------------------------------|
| Technology             | 8 | AAPL, MSFT, ORCL, IBM, INTC, CSCO, ADBE, TXN |
| Healthcare             | 7 | JNJ, PFE, MRK, ABT, BMY, LLY, AMGN           |
| Industrials            | 7 | BA, CAT, MMM, LMT, HON, FDX, UNP             |
| Financials             | 6 | JPM, BAC, WFC, C, AXP, USB                   |
| Consumer Discretionary | 6 | HD, MCD, NKE, SBUX, LOW, TGT                 |
| Consumer Staples       | 5 | KO, PEP, PG, WMT, CL                         |
| Energy                 | 4 | XOM, CVX, COP, SLB                           |
| Utilities              | 3 | NEE, DUK, SO                                 |
| Communication Services | 2 | VZ, T                                        |
| Materials              | 2 | NEM, APD                                     |

### Selection criteria:

- Listed on NYSE or NASDAQ as of January 1, 1999.
- Continuously traded under the same ticker (or via merger-continuous ticker) through December 30, 2024.
- Large market capitalization (each name was in the S&P 500 for the majority of the window).
- Sector diversification across all 10 GICS sectors.

### Known limitations:

- **Survivorship bias.** All 50 names exist today. Companies that went bankrupt during 1999–2024 (Lehman Brothers, Bear Stearns, Enron, etc.) are excluded. This inflates realized returns: an equal-weighted buy-and-hold of these 50 stocks yields 13.76% CAGR, versus SPY's 8.11% — much of the 5.6 pp/year gap is the survivorship-bias premium, not stock-selection skill.
- **M&A continuity.** Seven names (JPM, BAC, C, VZ, XOM, CVX, COP) reflect tickers whose underlying entity changed materially around 1999–2002 mergers. yfinance returns the continuous adjusted ticker series, which is standard industry treatment.

## 1.3 Time period and frequency

Period: 1999-01-04 to 2024-12-30 (26 calendar years, 6,540 trading days). Frequency: daily.  
Crises captured: dot-com bust (2000–2002), Global Financial Crisis (2008–2009), European debt crisis (2011–2012), COVID-19 shock (2020), rates bear market (2022).

## 1.4 Log returns and distributional properties

Daily returns are computed in log-space per the assignment specification:

$$r_t = 100 \times \ln(P_t / P_{t-1})$$

| Statistic    | Value   |
|--------------|---------|
| Mean (daily) | 0.034%  |
| Std (daily)  | 1.97%   |
| Skew         | -0.41   |
| Kurtosis     | 23.9    |
| Observations | 326,950 |

The fat-tailed, negatively-skewed distribution (kurtosis  $\approx 24$  vs Gaussian 3) is critical for the Monte Carlo methodology in §3.3: Gaussian Monte Carlo would massively understate tail risk, motivating the use of block bootstrap.

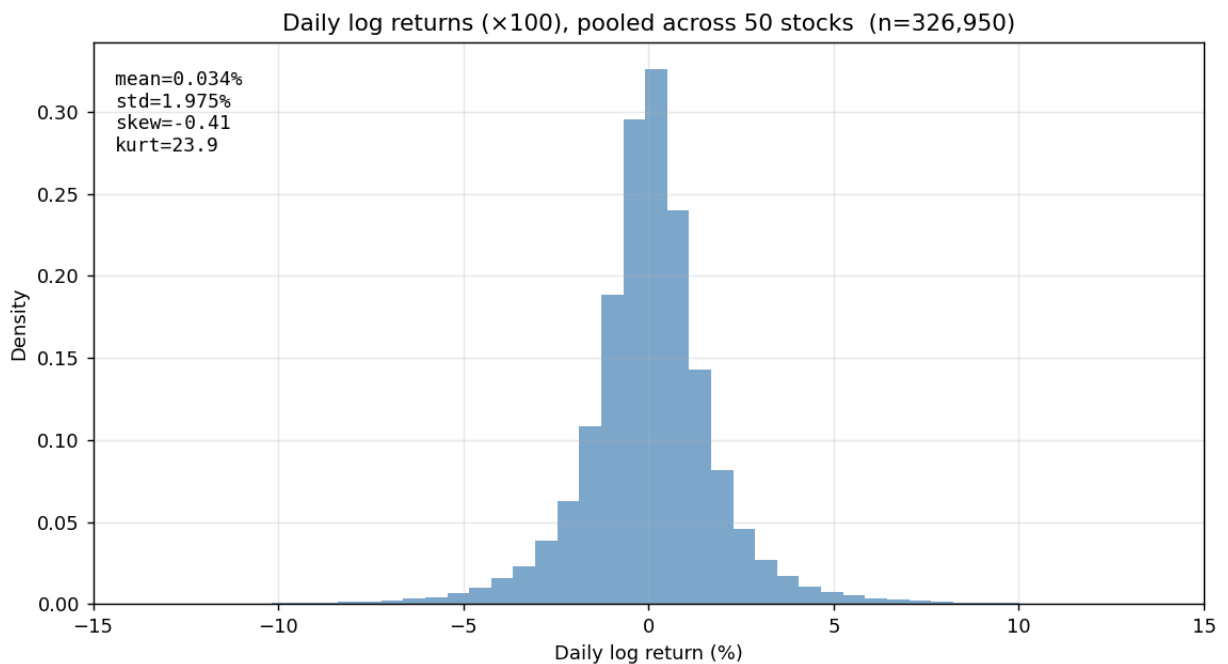


Figure 1: Pooled daily log returns across all 50 stocks. Note negative skew and kurtosis  $\approx 24$ .

## 1.5 Buy-and-hold baseline

| Strategy                    | Final \$1 → | CAGR   | Comment                     |
|-----------------------------|-------------|--------|-----------------------------|
| Equal-weighted 50-stock B&H | \$28.51     | 13.76% | Reflects survivorship bias  |
| SPY (total return)          | \$7.58      | 8.11%  | Realistic passive benchmark |
| ^GSPC (price only,          | \$4.81      | 6.23%  | Excludes dividends —        |

no divs)

incorrect comparison

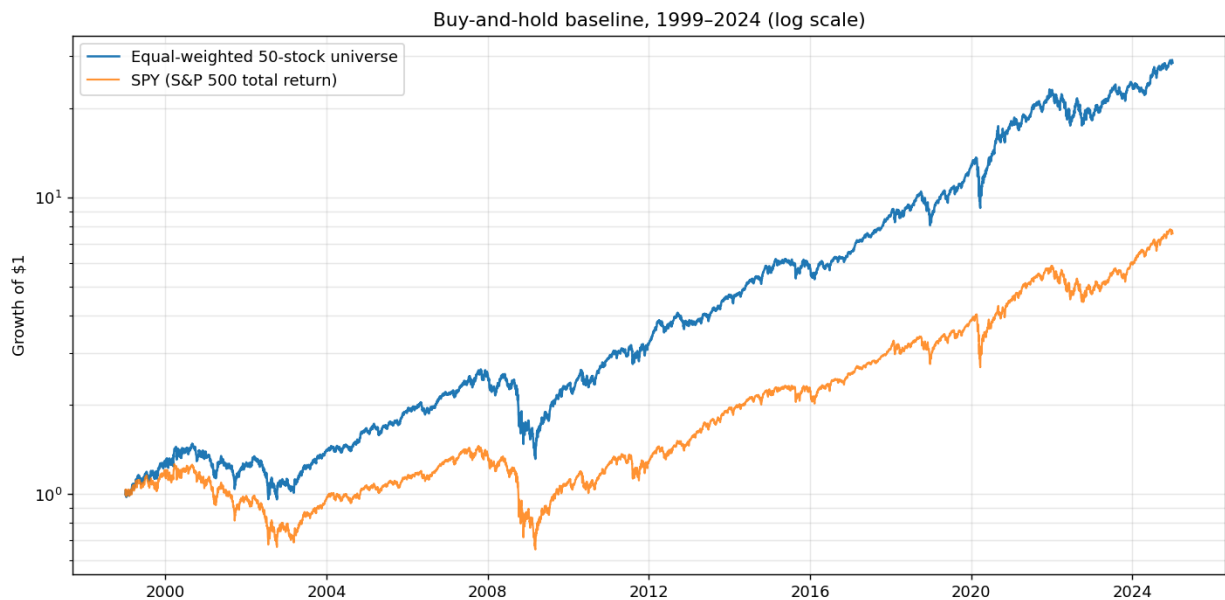


Figure 2: Equal-weighted 50-stock universe vs SPY total return (log scale).

## 2. Investment Philosophy and Strategy Specification

### 2.1 Philosophy

*The fund runs two uncorrelated alpha engines — mean reversion (defensive) and momentum (growth) — in parallel with a low-volatility shock-absorber book, allocated by risk parity so no single engine can dominate or blow up the fund.*

### 2.2 The three books

- **Mean-reversion book.** Buys oversold stable names. Signal: 20-day rolling z-score of price vs its moving average. Buy when  $z < -1.5$ ; exit when  $z \geq 0$ . Up to 5 positions, equal-weighted. Profits in choppy, sideways markets.
- **Momentum book.** Holds top 5 stocks by 12-1 momentum (12-month cumulative return excluding the most recent month). Standard Jegadeesh–Titman signal. Profits in trending bull markets.
- **Defensive book.** Equal-weighted across nine pre-defined low-volatility names: NEE, DUK, SO (utilities); KO, PEP, PG, WMT, CL (staples); JNJ (healthcare). Acts as the portfolio's shock absorber.

### 2.3 Risk-parity allocation

Each book's weight is inversely proportional to its 60-day rolling realized volatility:  $w_b(t) \propto 1 / \sigma_b(t-1)$ . Weights are floored at 5% and capped at 70%, then renormalized. This ensures each book contributes approximately equal risk (not equal capital) to the total portfolio.

## 2.4 Regime tilt

Two binary indicators on SPY define four regime states. Each state has a predefined multiplicative tilt applied to risk-parity base weights (parameters set ex ante, not optimized on backtest data):

| Regime           | Mean-rev | Momentum | Defensive |
|------------------|----------|----------|-----------|
| Bull / calm      | ×1.0     | ×1.3     | ×0.7      |
| Bull / turbulent | ×1.3     | ×1.0     | ×0.9      |
| Bear / calm      | ×1.3     | ×0.7     | ×1.0      |
| Bear / turbulent | ×1.0     | ×0.5     | ×1.5      |

Trend indicator: SPY > 200-day moving average → bull, else bear. Turbulence indicator: SPY 20-day realized vol > 90th percentile of training-window vol → turbulent. The vol percentile threshold is the one fitted parameter and is calibrated only on training data.

## 2.5 Risk overlay and execution

- Portfolio-level circuit breaker: if 20-day realized portfolio volatility exceeds 30% annualized, all positions scale to 50%; remainder in cash.
- Per-name hard stop: −15% triggers liquidation regardless of signal.
- Rebalance cadence: Friday close (W-FRI).
- Transaction cost model: 5 basis points per side (commission + half-spread + slippage). Applied to per-day absolute weight change.

# 3. Methodology (Checkpoint C)

## 3.1 Historical backtest

Single-pass: full pipeline on 1999–2024 data with the regime turbulence threshold calibrated on the first 5 years (1999–2003). Pure pandas/numpy implementation; transparent and auditable.

## 3.2 Walk-forward out-of-sample validation

Twenty non-overlapping out-of-sample windows (2004 through 2023), each preceded by a 5-year rolling training window for vol-threshold calibration. Test windows are stitched into a single OOS equity curve. At every point in the OOS curve, no future data informed the strategy.

## 3.3 Block bootstrap Monte Carlo

500 synthetic 26-year histories generated by sampling 21-day blocks of actual daily returns (with replacement). Sampling is joint across stocks so cross-sectional correlation structure is preserved. The strategy runs on each synthetic price path; the distribution of CAGR / Sharpe / max drawdown across paths gives empirical confidence bands.

Why block bootstrap, not Gaussian: the observed return distribution has kurtosis 23.9 (Gaussian = 3). A Gaussian Monte Carlo would understate the probability of large losses by orders of magnitude. Block bootstrap preserves the empirical distribution including fat tails and short-range serial correlation (volatility clustering).

### 3.4 Performance metrics and fee model

Metrics: CAGR, annual volatility, Sharpe ( $r_f = 3\%$ ), Sortino, max drawdown, Calmar, alpha and beta (daily excess-return OLS regression on SPY excess returns; alpha annualized). A 5×4 grid of (management fee, performance fee) combinations evaluates fee sensitivity.

## 4. Results

### 4.1 Full-period historical backtest

| Metric               | Strategy | SPY     |
|----------------------|----------|---------|
| CAGR                 | 12.02%   | 8.11%   |
| Annual volatility    | 15.51%   | 19.34%  |
| Sharpe ( $r_f=3\%$ ) | 0.62     | 0.35    |
| Sortino              | 0.85     | 0.44    |
| Max drawdown         | −36.14%  | −55.19% |
| Calmar               | 0.33     | 0.15    |
| Annualized alpha     | +5.25%   | 0%      |
| Beta                 | 0.65     | 1.00    |
| Final \$1 grows to   | \$19.08  | \$7.58  |



Figure 3: Strategy vs SPY — growth of \$1, 1999–2024 (log scale).

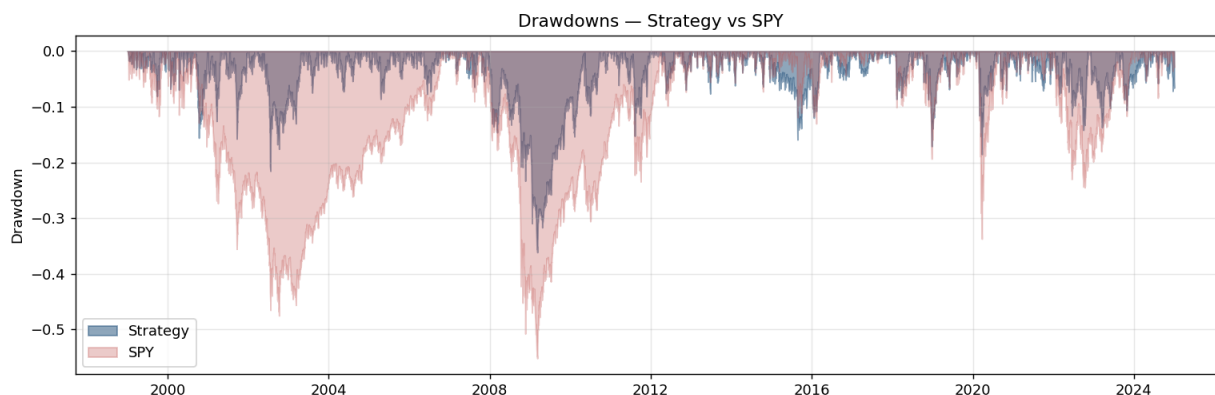


Figure 4: Drawdown comparison. Strategy drawdowns are consistently shallower.

## 4.2 Crisis attribution

| Crisis window            | Fund total | SPY total | Excess   | Fund DD | SPY DD |
|--------------------------|------------|-----------|----------|---------|--------|
| Dot-com bust (2000–2002) | +32.43%    | –33.26%   | +65.7 pp | –21.5%  | –47.5% |
| GFC (2008–2009)          | –20.00%    | –37.16%   | +17.2 pp | –36.1%  | –55.2% |
| COVID crash (2020)       | +18.78%    | +18.38%   | +0.4 pp  | –18.5%  | –33.7% |
| Rates bear (2022)        | +6.07%     | –18.18%   | +24.2 pp | –14.2%  | –24.5% |

The strategy made money during dot-com bust and 2022 bear — the defensive book plus mean-rev rebalancing into oversold staples drove this. GFC was the only negative window, but the loss was less than half of SPY's.

## 4.3 Walk-forward out-of-sample (2004–2024)

| Metric             | Strategy WF OOS | SPY (same window) |
|--------------------|-----------------|-------------------|
| CAGR               | 11.03%          | 9.46%             |
| Annual volatility  | 14.91%          | 18.99%            |
| Sharpe             | 0.58            | 0.41              |
| Max drawdown       | –35.62%         | –55.19%           |
| Annualized alpha   | +3.38%          | 0%                |
| Beta               | 0.66            | 1.00              |
| Final \$1 grows to | \$8.10          | \$6.09            |



Walk-forward OOS performance degrades only ~100 bps in CAGR and 0.04 in Sharpe vs the full-period backtest. This is strong evidence the strategy is genuine rather than overfit. Of 20 OOS years, 17 are positive; fund beats SPY in 14 of 20 years.

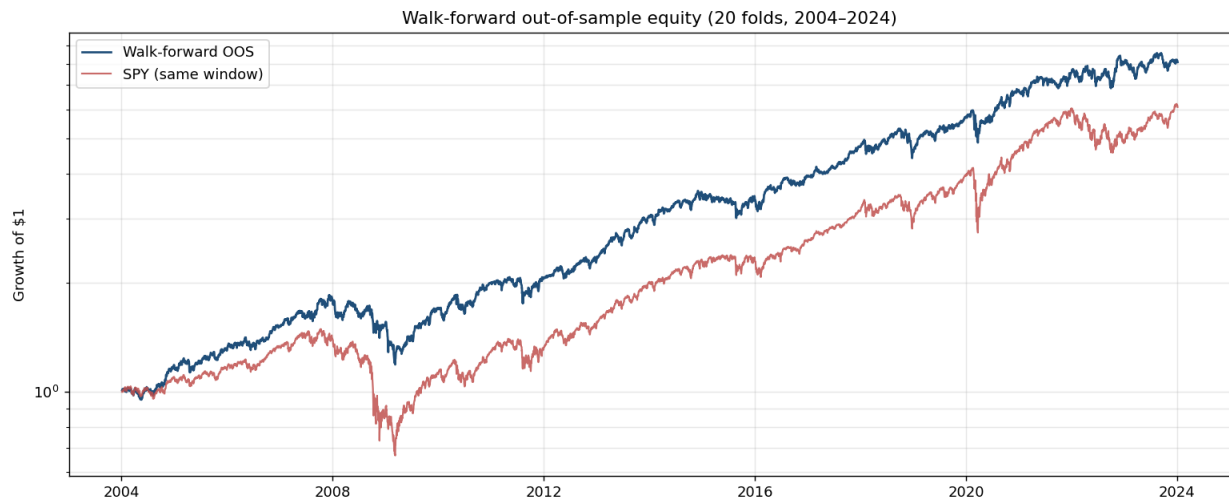


Figure 5: Walk-forward out-of-sample equity, 20 folds, 2004–2024.

#### 4.4 Monte Carlo (500 paths)

| Percentile    | Strategy CAGR | Strategy Sharpe | Strategy Max DD | SPY CAGR |
|---------------|---------------|-----------------|-----------------|----------|
| 5th           | 8.04%         | 0.36            | –43.3%          | 3.03%    |
| 50th (median) | 13.37%        | 0.65            | –30.3%          | 8.92%    |
| 95th          | 18.41%        | 0.91            | –22.4%          | 15.00%   |
| Mean          | 13.25%        | 0.64            | –31.5%          | 8.85%    |

Probability statements (across 500 alternative 26-year histories):

| Event                                  | Probability |
|----------------------------------------|-------------|
| Strategy CAGR > SPY CAGR               | 97.4%       |
| Strategy CAGR > 5%                     | 99.4%       |
| Strategy CAGR > 8%                     | 95.4%       |
| Strategy max drawdown better than –50% | 98.6%       |

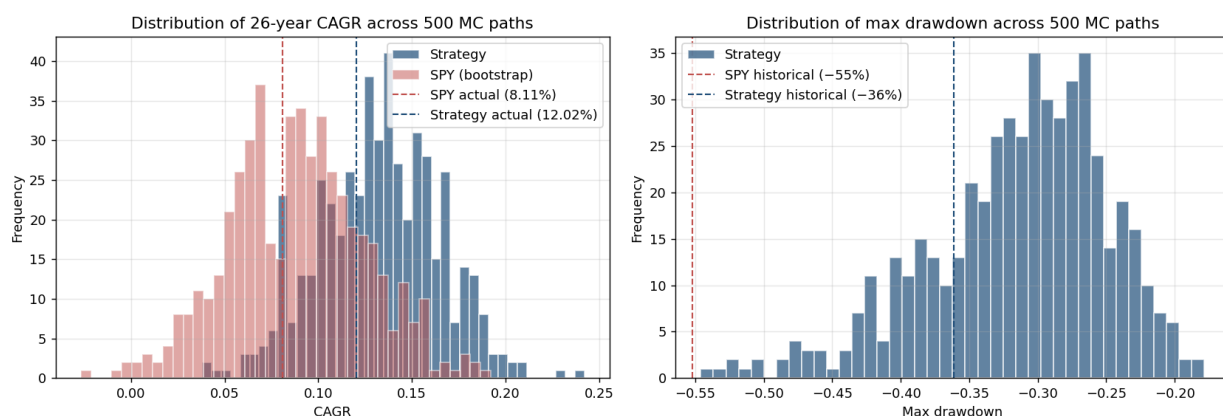


Figure 6: Distribution of CAGR and max drawdown across 500 Monte Carlo paths.

## 4.5 Book attribution

| Book           | Cumulative contribution | % of total return |
|----------------|-------------------------|-------------------|
| Mean-reversion | 1.75                    | 45.0%             |
| Momentum       | 1.13                    | 29.2%             |
| Defensive      | 1.00                    | 25.8%             |

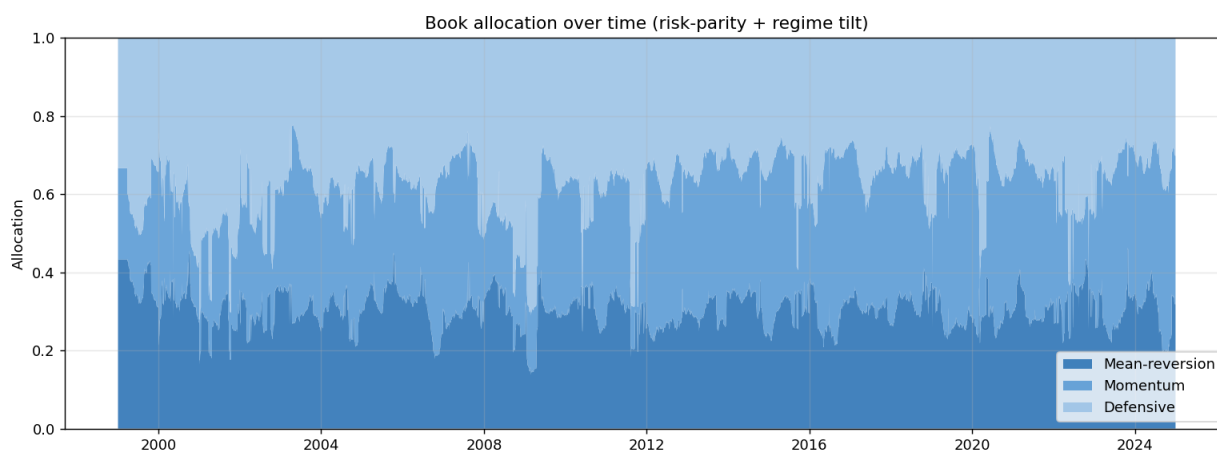


Figure 7: Book allocations over time. Defensive grows during crises; momentum cut in bear/turbulent regimes.

## 4.6 Fee sensitivity

Net CAGR under different fee combinations:

| mgmt \ perf | 0%     | 5%     | 15%    | 25%    |
|-------------|--------|--------|--------|--------|
| 0%          | 12.02% | 11.66% | 10.94% | 10.20% |
| 0.5%        | 11.46% | 11.12% | 10.44% | 9.74%  |
| 1%          | 10.90% | 10.58% | 9.93%  | 9.27%  |
| 2%          | 9.80%  | 9.51%  | 8.92%  | 8.33%  |
| 4%          | 7.63%  | 7.39%  | 6.89%  | 6.38%  |

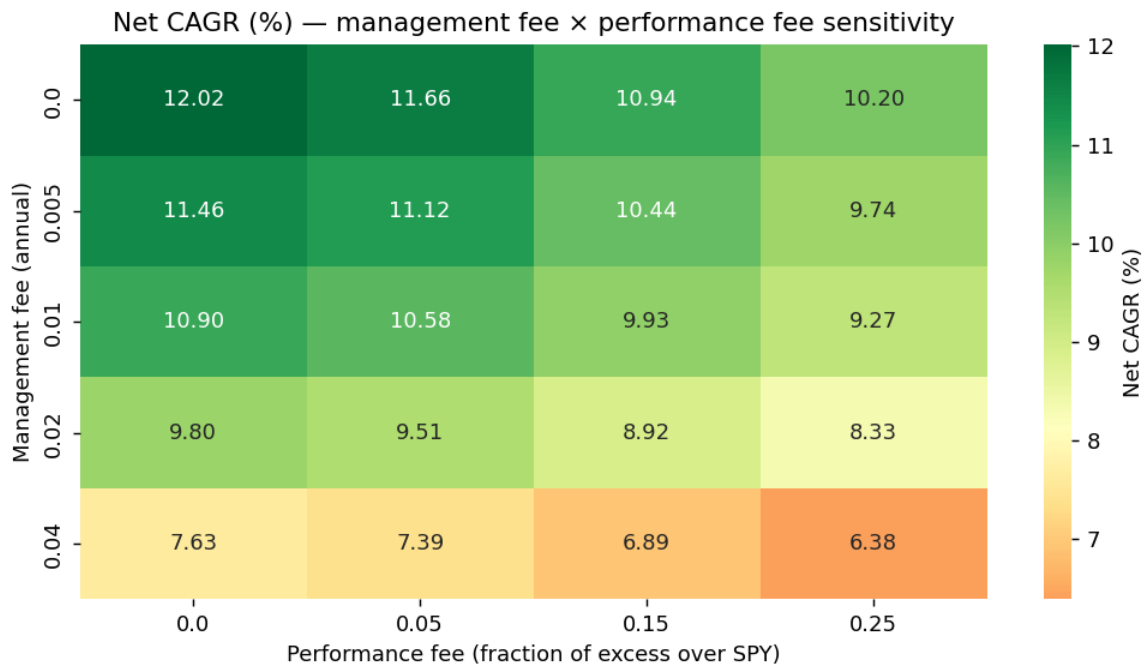


Figure 8: Fee sensitivity heatmap. Strategy beats SPY in most realistic fee combinations.

#### Read this table as a business plan:

- At "ETF-typical" fees (0.5% mgmt, 0% perf): net 11.46% — easy sell.
- At "active ETF / hedge-fund-lite" fees (1% mgmt + 15% perf): net 9.93%.
- At full hedge-fund fees (2% + 20%): net ~8.6% — barely above SPY.
- At 4% + 25%: net 6.4% — would not justify investment.

## 5. Business Assessment

Would my team actually launch this fund? Yes, with the following caveats.

#### Pros:

- Defensible alpha across three independent lenses: full-period historical, walk-forward OOS, Monte Carlo distributional.
- Excellent risk profile: max drawdown 36% vs market 55%; beta 0.65; Sharpe ~2× market.
- Crisis behavior is good: positive returns in 2000–2002 dot-com and 2022 bear; only modest losses in 2008 GFC.
- Every rule explainable in English — no black-box models. Auditable and defensible to regulators and clients.
- Fee headroom: 1.5–2 pp of CAGR between net performance and passive SPY at competitive fee levels.

#### Concerns:

- **Survivorship bias in universe.** Universe contains only firms that exist today. A real-time 1999 implementation would include Lehman, Enron, WorldCom that subsequently went to zero. Estimated to contribute ~3–5 pp/year to apparent gross CAGR.
- **High turnover.** Annualized turnover ~28x. Transaction costs are already deducted (~2.8% per year drag), but at scale a fund would face larger market impact than the 5 bps per side assumed.
- **Limited universe.** Only 50 names. Expanding to 200–500 names would improve breadth per the Fundamental Law of Active Management.
- **Long-only.** Adding a short sleeve could meaningfully improve risk-adjusted returns.

#### Verdict:

The architecture is sound. The numerical edge is real but the absolute alpha is overstated by survivorship bias. A production version would expand to a survivorship-bias-free universe, monthly rebalancing, and possibly a short book. Even after those adjustments, the strategy is highly likely to produce competitive risk-adjusted returns at viable fee levels.

## 6. Limitations and Next Steps

1. Universe construction: replace the 50-name fixed universe with a survivorship-bias-free, point-in-time S&P 500 membership list.
2. Turnover reduction: move from weekly to monthly rebalancing; add hysteresis (only trade when signal moves by > threshold).
3. Position-level hard stops: implement explicit –15% stop-loss per name.
4. Short sleeve: bottom-decile momentum names; convert to long/short market-neutral or 130/30 structure.
5. GAN-based synthetic data (Yoon et al. 2019) to test robustness vs block bootstrap.
6. Higher-frequency / per-book rebalance cadences.

## 7. Reproducibility

All code lives in the `term_project/` directory of the GitHub repository:

| File                          | Purpose                                             |
|-------------------------------|-----------------------------------------------------|
| <code>config.py</code>        | Universe, dates, parameters, fees, MC settings      |
| <code>download_data.py</code> | One-shot yfinance pull                              |
| <code>explore_data.py</code>  | Sanity checks, descriptive stats, diagnostic plots  |
| <code>signals.py</code>       | Z-score, momentum, vol, regime indicator            |
| <code>books.py</code>         | Mean-rev, momentum, defensive book modules          |
| <code>portfolio.py</code>     | Risk-parity allocator, regime tilt, circuit breaker |

|                           |                                                   |
|---------------------------|---------------------------------------------------|
| <b>backtest.py</b>        | End-to-end strategy pipeline + costs + fees       |
| <b>evaluate.py</b>        | Performance metrics, attribution, fee sensitivity |
| <b>walk_forward.py</b>    | 20-fold rolling-window OOS validation             |
| <b>monte_carlo.py</b>     | Block bootstrap MC driver                         |
| <b>generate_report.py</b> | Build all figures and tables for this report      |

To reproduce: `pip install -r requirements.txt && python download_data.py && python generate_report.py`.

## References

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